

Domain-Driven Design Context Map + Business Domains

Institute Management System (IMS)

Version: 1.2

Scope: Single-client implementation

Excluded for now: Tenant Setup, SaaS Subscription Management, CMS / Website Builder

Primary Business Focus: Coaching center, skill training institute, individual training, corporate training operations, and walk-in course completion programs

1. Purpose

This document defines the Domain-Driven Design (DDD) structure for the IMS platform. It identifies the major business domains, bounded contexts, ownership boundaries, integrations, lifecycle events, and relationships between modules.

The goal is to establish a clean business architecture that supports:

- Database design
 - API design
 - Next.js monorepo modular architecture
 - Future scalability
 - Clear domain ownership
 - Phased implementation planning
-

2. Domain Classification

The IMS platform is organized into Core Domains, Supporting Domains, and Generic Domains.

2.1 Core Domains

These domains directly drive business value and differentiate the institute.

1. Lead & Inquiry Management
 2. Admission & Enrollment Management
 3. Course, Batch & Completion Rule Management
 4. Fee & Finance Management
 5. Certificate Management
 6. Corporate Training Management
 7. Walk-In Enrollment & Completion Flow
-

2.2 Supporting Domains

These domains support operational execution.

1. Organization Management
 2. Faculty / Trainer Management
 3. Scheduling & Timetable Management
 4. Attendance Management
 5. Exam, Result & Completion Management
 6. Communication Management
 7. Document Management
-

2.3 Generic Domains

These domains provide platform capabilities.

1. Identity & Access Management
 2. Reporting & Analytics
 3. Audit & Compliance
 4. Notification Infrastructure
 5. AI Intelligence (Future Phase)
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3. High-Level Business Domains

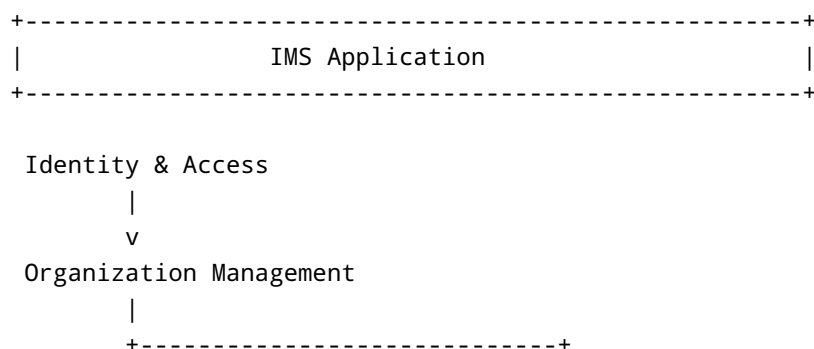
The IMS system is divided into the following bounded contexts:

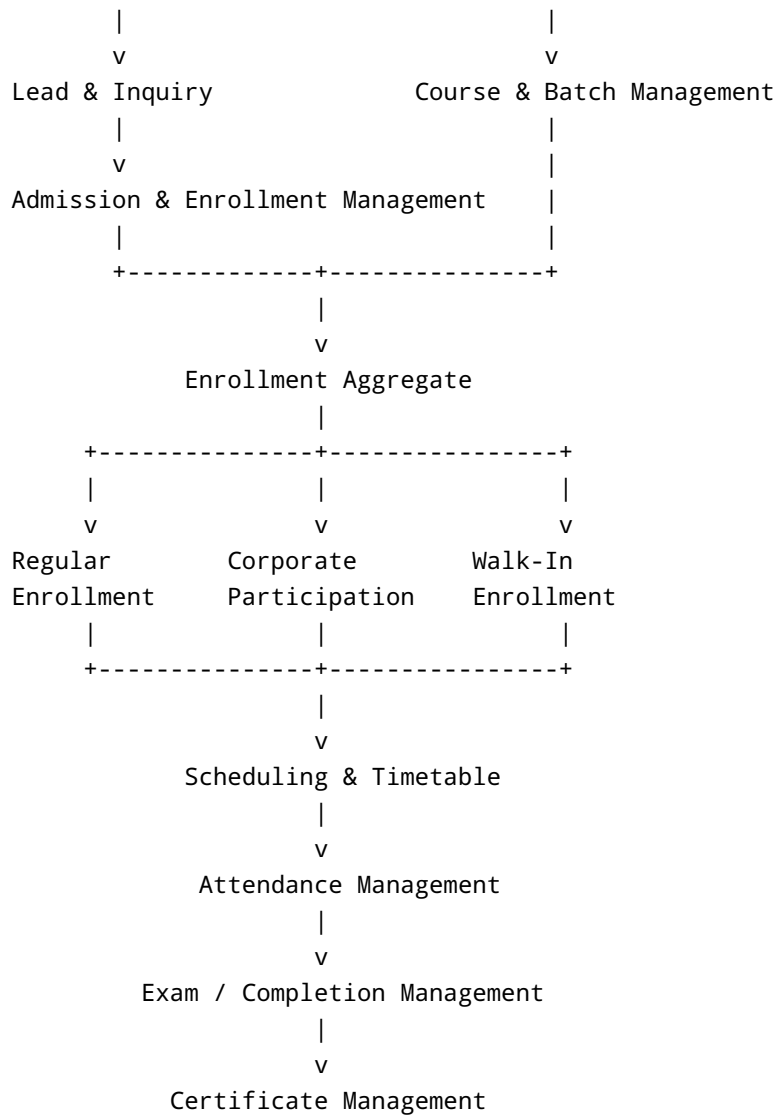
1. Identity & Access Management
 2. Organization Management
 3. Lead & Inquiry Management
 4. Admission & Enrollment Management
 5. Corporate Training Management
 6. Course & Batch Management
 7. Walk-In Enrollment & Completion Flow
 8. Scheduling & Timetable Management
 9. Attendance Management
 10. Fee & Finance Management
 11. Faculty / Trainer Management
 12. Exam, Result & Completion Management
 13. Certificate Management
 14. Communication Management
 15. Document Management
 16. Reporting & Analytics
 17. Audit & Compliance
 18. AI Intelligence – Future Phase
-

4. Phase-Based Context Planning

Context	Phase
Identity & Access Management	Phase 1
Organization Management	Phase 1
Lead & Inquiry Management	Phase 1
Admission & Enrollment Management	Phase 1
Course & Batch Management	Phase 1
Fee & Finance Management (Manual Payments)	Phase 1
Faculty / Trainer Management	Phase 1
Attendance Management	Phase 1
Exam, Result & Completion Management	Phase 1
Certificate Management	Phase 1
Walk-In Enrollment & Completion Flow	Phase 1
Document Management	Phase 1
Communication Management	Phase 2
Corporate Training Management	Phase 2
Trainer Payment Tracking	Phase 2
Advanced Reporting & Analytics	Phase 2
Online Payment Gateway	Final Phase
Payroll Processing	Final Phase
AI Intelligence	Final Phase

5. DDD Context Map Overview





Corporate Training Management

- |
- +--> Corporate Accounts
- +--> Corporate Contracts
- +--> Corporate Programs
- +--> Corporate Participants

Fee & Finance Management

- |
- +--> Manual Payments
- +--> Discounts
- +--> Refunds
- +--> Receipts
- +--> Due Tracking

Faculty / Trainer Management

- |
- +--> Trainer Profiles
- +--> Qualifications

```
+--> Documents
+--> Availability
+--> Assignments
+--> Payment Tracking
```

Communication Management

```
|
+--> SMS
+--> WhatsApp
+--> Email
+--> Notification History
```

Document Management

```
|
+--> Student Documents
+--> Trainer Documents
+--> Verification Workflow
```

Reporting & Analytics

```
|
+--> Operational Reports
+--> Management Dashboard
+--> Future AI Insights
```

Audit & Compliance

```
|
+--> User Actions
+--> Finance Changes
+--> Attendance Changes
+--> Approval History
```

6. Central Enrollment Aggregate

Business Purpose

Enrollment is the central aggregate that connects students, participants, courses, batches, finance, completion, and certificates.

This aggregate provides a unified lifecycle regardless of whether the learner is:

- Regular student
 - Corporate participant
 - Walk-in participant
-

Aggregate Root

Enrollment

Core Fields

Enrollment Number
Enrollment Type
Student Reference
Corporate Participant Reference
Course
Batch / Session
Branch
Fee Plan
Enrollment Status
Completion Status
Certificate Eligibility Status

Enrollment Types

Regular
Corporate
WalkIn

Enrollment Statuses

Draft
Confirmed
Active
Completed
Cancelled
Dropped
Certificate Issued

Key Benefits

- Single enrollment lifecycle
- Shared completion logic
- Shared certificate eligibility logic

- Shared finance integration
 - Reduced duplication across domains
-

7. Bounded Contexts

7.1 Identity & Access Context

Phase

Phase 1

Business Purpose

Controls authentication, authorization, roles, permissions, and secure access.

Key Responsibilities

- User login
- Password management
- Dynamic role creation
- Permission assignment
- Menu-level access
- Report-level access
- Action-level access

Core Entities

- User
- Role
- Permission
- Menu
- UserRole
- RolePermission
- AccessPolicy

Important Rules

- Roles must not be hardcoded.
 - Permissions should be configurable.
 - Access should be controlled by action, menu, and report.
 - Student, trainer, accountant, counselor, and branch manager access must be separated.
 - Branch managers should only access assigned branch data.
 - Counselors should primarily access assigned leads, with optional permission to view all branch leads and assigned counselor information.
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7.2 Organization Management Context

Phase

Phase 1

Business Purpose

Defines the organizational structure of the institute.

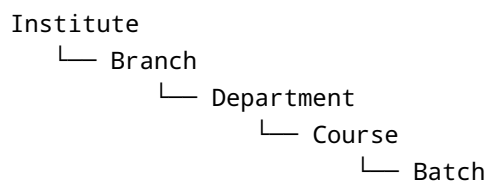
Key Responsibilities

- Branch management
- Department management
- Classroom management

Core Entities

- Branch
- Department
- Classroom

Business Hierarchy



7.3 Lead & Inquiry Management Context

Phase

Phase 1

Business Purpose

Manages inquiries, campaigns, lead pipelines, follow-ups, and conversion.

Key Responsibilities

- Capture inquiries
- Track lead source
- Manage lead pipeline
- Assign counselor
- Track follow-ups
- Convert lead to admission

Important Rules

- Lead sources must be configurable.
 - Follow-up reminders should generate notifications.
 - Won leads should move into Admission & Enrollment.
 - Lost reasons must be captured.
-

7.4 Admission & Enrollment Management Context

Phase

Phase 1

Business Purpose

Manages student registration, admissions, enrollment lifecycle, student identity management, and student records.

Key Responsibilities

Admission

- Register student/customer
- Create student profile
- Manage student identity information
- Generate student IDs
- Generate student ID cards

Enrollment

- Course enrollment
- Batch enrollment
- Walk-in enrollment
- Corporate-linked enrollment
- Enrollment lifecycle tracking

Core Entities

- Student
- Admission
- Enrollment
- StudentIdentity
- StudentIDCard

Important Rules

Admission

Admission represents becoming a registered student/customer of the institute.

Enrollment

Enrollment represents joining:

- Course
- Batch
- Corporate program
- Walk-in session

A single student may have multiple enrollments over time.

7.5 Corporate Training Management Context

Phase

Phase 2

Business Purpose

Manages corporate customers, contracts, programs, participants, and corporate training delivery.

Core Entities

- CorporateAccount
- CorporateContactPerson
- CorporateContract
- CorporateTrainingProgram
- CorporateBatch
- CorporateParticipant

Optional Entity

LinkedStudentProfile

Important Rules

Corporate employees should not automatically become students.

Default lifecycle:

Corporate Participant
↓
(Optional)
Linked Student Profile

A participant should only be linked to a Student profile when:

- Login access is required
 - Individual lifecycle tracking is required
 - Future enrollments are expected
 - Certificate history must be retained
 - Individual payments are required
-

7.6 Course & Batch Management Context

Phase

Phase 1

Business Purpose

Manages courses, pricing, batches, trainers, capacity, waiting lists, and completion rules.

Core Entities

- Course
- CourseType
- CoursePricing
- CourseCompletionRule
- Batch
- BatchCapacity
- BatchTrainer
- WaitingList

Important Rules

- Same course can have different pricing by branch, batch type, and customer type.
- Completion rules are defined at course level.
- Completion requirements are inherited by batches.
- Waiting lists are required when capacity is reached.

Course Completion Types

Attendance Based
Completion Only
Exam Required
Exam + Attendance
Manual Approval
Walk-In Completion

7.7 Walk-In Enrollment & Completion Flow

Phase

Phase 1

Business Purpose

Provides a specialized enrollment path for learners who complete training immediately or within a short duration without joining a traditional scheduled batch.

Domain Position

Walk-In is not a standalone admission domain.

It is a specialized enrollment and completion flow built on top of:

- Course Management
- Enrollment Management
- Finance
- Completion Management
- Certificate Management

Enrollment Type

WalkIn

Key Responsibilities

- Same-day enrollment
- Trainer assignment
- Completion validation
- Certificate eligibility validation
- Immediate certificate generation

Important Rules

- Course must explicitly allow walk-in completion.
 - Same-day enrollment is allowed.
 - Same-day completion is allowed.
 - Completion rules are inherited from the course.
 - Payment completion may be mandatory before certificate generation.
 - Certificate eligibility must be validated before issuance.
-

7.8 Scheduling & Timetable Context

Phase

Phase 1

Business Purpose

Manages schedules, trainer allocation, classroom allocation, and conflict detection.

Important Rules

System must prevent:

Trainer double booking
Classroom double booking
Batch overlap

7.9 Attendance Management Context

Phase

Phase 1

No major changes.

7.10 Fee & Finance Management Context

Phase

Phase 1

Business Purpose

Manages fees, manual payments, receipts, discounts, refunds, and due tracking.

Phase 1 Scope

Manual Payment Entry
Receipt Generation
Fee Due Tracking
Discounts
Refunds

Reference Numbers
Payment Remarks

Supported Payment Modes

Cash
Bank Transfer
Card
Manual Online Transfer
Cheque

Receipt Formats

Tax Invoice
Simplified Tax Invoice
Receipt Voucher
Advance Receipt
Corporate Invoice
Refund Receipt

Final Phase Scope

Payment Gateway
Online Payment Links
Webhook Confirmation
Auto Reconciliation
Failed Payment Handling
Gateway Refund Integration

7.11 Faculty / Trainer Management Context

Phase

Phase 1

Business Purpose

Manages trainer information, qualifications, assignments, availability, and trainer payment tracking.

Core Entities

- Trainer
- TrainerQualification
- TrainerDocument
- TrainerAvailability

- TrainerAssignment
- TrainerPayment

Responsibilities

- Trainer profile management
- Qualification tracking
- Document management
- Availability management
- Batch assignment
- Trainer payment tracking

Excluded from Phase 1

Full payroll processing is intentionally excluded.

Payroll Management (Final Phase)

Future bounded context:

Payroll Management

Future responsibilities:

- Salary structures
 - Payroll calculations
 - Deductions
 - Payslips
 - Payroll approvals
 - Statutory payroll compliance
-

7.12 Exam, Result & Completion Context

Phase

Phase 1

Business Purpose

Manages exams, results, completion validation, and completion approvals.

Completion Approval Workflow

Trainer Recommendation
↓
Academic Coordinator Review

↓
Branch Manager Approval
↓
Completion Approved

7.13 Certificate Management Context

Phase

Phase 1

Business Purpose

Manages certificate generation, issuance, verification, and lifecycle management.

Important Rules

Certificate eligibility may depend on:

Attendance Percentage
Course Completion
Exam Pass
Manual Approval
Walk-In Completion
Payment Completion

7.14 Communication Management Context

Phase

Phase 2

No major changes.

7.15 Document Management Context

Phase

Phase 1

Recommended Initial Document Types

Civil ID
Passport Copy
Visa Copy
Passport Photo
Educational Certificate
Employment Letter
Sponsor Letter
Corporate Authorization Letter
Trainer Qualification Certificate
Trainer Contract

7.16 Reporting & Analytics Context

Phase

Phase 2

No major changes.

7.17 Audit & Compliance Context

Phase

Phase 1

No major changes.

7.18 AI Intelligence Context – Future Phase

Phase

Final Phase

No major changes.

8. Domain Events

Shared Lifecycle Events

```
EnrollmentCreated  
EnrollmentConfirmed  
ManualPaymentRecorded  
CourseCompleted  
CertificateEligibilityApproved  
CertificateGenerated  
CorporateParticipantRegistered  
WalkInCompletionApproved  
TrainerPaymentRecorded
```

These events should be used across bounded contexts instead of creating isolated module-specific events wherever possible.

9. Context Integration Rules

Enrollment Lifecycle

```
Admission  
  ↓  
Enrollment Created  
  ↓  
Enrollment Confirmed  
  ↓  
Training Delivery  
  ↓  
Completion Validation  
  ↓  
Certificate Eligibility  
  ↓  
Certificate Generation
```

Walk-In Completion Flow

```
Walk-In Enrollment  
  ↓  
Completion Validation  
  ↓  
Payment Validation
```

↓
Certificate Eligibility
↓
Certificate Generation

Corporate Training Flow

Corporate Account
↓
Corporate Contract
↓
Corporate Program
↓
Corporate Participant
↓
(Optional)
Linked Student Profile
↓
Enrollment

10. Architecture Approach

Recommended Architecture

Use a **Next.js Monorepo Modular Architecture** instead of microservices.

Suggested Monorepo Structure

```
ims-monorepo
|
|— apps
|   |— admin-portal
|   |— student-portal
|   |— trainer-portal
|   |— public-verification
|
|— packages
|   |— identity-access
|   |— organization
|   |— crm-leads
|   |— admissions-enrollment
|   |— corporate-training
|   |— courses-batches
```

```

├─ walkin-enrollment
├─ scheduling
├─ attendance
├─ finance
├─ trainer-management
├─ exams-completion
├─ certificates
├─ communication
├─ documents
├─ reporting
├─ audit
├─ shared-ui
├─ shared-types
├─ shared-utils
└─ infrastructure
    ├─ database
    ├─ auth
    ├─ storage
    └─ integrations

```

11. Data Ownership Rules

Data	Owning Context
User, Role, Permission	Identity & Access
Branch, Department, Classroom	Organization
Lead, Follow-up, Campaign	Lead Management
Student, Admission, Enrollment, Student ID Card	Admission & Enrollment
Corporate Account, Contract, Participant	Corporate Training
Course, Pricing, Completion Rules	Course Management
Batch, Waiting List	Course & Batch Management
Schedule, Session	Scheduling
Attendance Record	Attendance
Fee, Payment, Receipt, Refund	Finance
Trainer, Qualification, Availability	Trainer Management
Trainer Payment Tracking	Trainer Management
Exam, Result, Completion	Exam & Completion
Certificate	Certificate Management

Data	Owning Context
Template, Message Log	Communication
Document, Verification	Document Management
Audit Log, Approval	Audit & Compliance

12. Key Design Decisions

Decision 1

Use Dynamic RBAC instead of hardcoded roles.

Decision 2

Use a Next.js Monorepo Modular Architecture.

Decision 3

Keep Tenant Setup and CMS out of current scope.

Decision 4

Design the system so SaaS and CMS can be added later without redesign.

Decision 5

Collect AI-ready data from Day 1 even if AI features are built later.

Decision 6

Make student identity fields configurable for Oman and future country expansion.

Decision 7

Separate Admission from Enrollment as distinct business concepts.

Decision 8

Use Enrollment as the central aggregate across regular, corporate, and walk-in learning journeys.

Decision 9

Treat Walk-In as a specialized enrollment and completion flow rather than a standalone admission domain.

Decision 10

Corporate employees begin as Corporate Participants and become Students only when required.

Decision 11

Support manual payment entry in Phase 1 and defer payment gateway integration to the final phase.

Decision 12

Replace Faculty & Payroll with Faculty / Trainer Management in early phases.

Decision 13

Move full Payroll Management to the final implementation phase.

Decision 14

Define course completion rules centrally at the course level.

Decision 15

Provide configurable student ID card generation and printing.

13. Resolved Business Decisions

Question	Decision
Refund Approval Hierarchy	Finance → Branch Manager → Management
Payroll Processing	Deferred to Final Phase
Trainer Payment Tracking	Supported before Payroll
Course Completion Approval	Trainer → Academic Coordinator → Branch Manager
Branch Manager Visibility	Assigned branch only
Counselor Visibility	Assigned leads by default
Receipt Formats	Oman-standard tax invoice and receipt formats